

Fabric property cards

Lesson 5 · Test materials and joins · part 1 of 2

These are fictional specimen descriptions for an offline lesson. They describe these samples, not every fabric with the same name. Match a property to evidence; handle real samples only if you choose to.

A | Thin woven cotton sample: bends easily; its surface looks smooth; a freshly cut edge has loose threads. It barely stretches during this class's gentle check.

B | Felt sample: feels soft; looks thicker than sample A; bends; its cut edge stays together in this class's handling check. It barely stretches.

C | Denim sample: feels firmer and thicker than A; bends with more effort; its cut edge has loose threads. This sample has little stretch.

D | Knitted jersey sample: bends and stretches under a gentle pull, then moves back towards its starting shape. Its surface looks closely looped.

Teacher answers are in the [separate guide](#).

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E | Net sample: visible holes let the card behind show through; it bends easily. A small loose item could pass through its holes.

Property word bank: flexible = bends; stretchy = changes length under a gentle pull; opaque = cannot see through; translucent/see-through = some light or details show through; thick/thin = relative thickness; frays = cut threads loosen.

Match jobs: a soft card pocket that keeps its cut edge together; a display overlay that reveals the card behind; a fabric item that needs stretch. Use the evidence on these cards, and name any further check needed.

Teacher answers are in the [separate guide](#).

Three join cards | What changes?

Lesson 5 · Test materials and joins · part 1 of 2

Look at prepared samples or these written descriptions. The core practical makes glue and cold-tape samples; an adult-prepared stitched sample allows a third method to be inspected without pupil needle use.

GLUE: a school-approved cold fabric adhesive joins two overlapping fabric pieces. Use the amount and curing time on its label. A fresh wet sample is for drying, not immediate testing.

TAPE: cold, pressure-applied double-sided fabric tape joins an overlap when pressed as its label directs. Use a product approved for the actual material. Iron-on or heat-activated tape is not used in this pack.

STITCH: thread travels through both fabric layers to form a join. The display sample is prepared by an adult before the lesson; pupils may point to its thread path. No needle task is required.

Teacher answers are in the [separate guide](#).

Three join cards | What changes?

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Look at prepared samples or these written descriptions. The core practical makes glue and cold-tape samples; an adult-prepared stitched sample allows a third method to be inspected without pupil needle use.

Temporary paper display repair: speed may matter; approved cold tape can be considered after a quick trial on the actual surface.

Garment hem: consider the fabric, comfort, movement, washing and whether the join is temporary or lasting. The small classroom test cannot decide all of these.

Bag handle: it carries a load repeatedly. It needs an appropriate design, material, reinforcement and a test for its actual load; this lesson does not certify any handle as safe.

No method is always strongest, neatest or fastest for every material. A join can change when its overlap, adhesive, drying, stitches or test conditions change.

Teacher answers are in the [separate guide](#).

Join test | Sample data and fair comparison

Lesson 5 · Test materials and joins · part 1 of 2

These are invented class results, not tests carried out by the pupil or a guarantee about a product. All three methods were tested on the same type of felt and the same setup. Read the method before comparing.

SAMPLE SETUP: two felt strips, each 100 mm × 30 mm, overlap by 20 mm × 30 mm. Each specimen uses glue, cold fabric tape, or an adult-prepared line of six running stitches across the overlap. Adhesive specimens have cured as their labels require.

SAMPLE TEST: an adult supports one end and attaches a sealed 100 g test load to the other for 10 seconds, with the load just above a tray. No pupil hangs weights or pulls a join towards their face. The same load, time and setup are used for every specimen.

REPEAT 1: glue held; tape opened; stitch held.

REPEAT 2: glue held; tape held; stitch opened.

Teacher answers are in the [separate guide](#).

Join test | Sample data and fair comparison

Lesson 5 · Test materials and joins · part 2 of 2

These are invented class results, not tests carried out by the pupil or a guarantee about a product. All three methods were tested on the same type of felt and the same setup. Read the method before comparing.

REPEAT 3: glue held; tape opened; stitch held.

“Held” means the layers stayed joined for the full 10 seconds at this load. “Opened” means the overlap separated before 10 seconds. Record observations rather than making the result fit an expected winner.

For real work: pupils can make a fresh glue and cold-tape sample using adult-prepared strips, then label them and leave them to cure. Inspect adult-prepared samples or use these data. Do not test fresh glue just because lesson time is short.

Fair-test check: keep fabric, strip size, overlap, load, time and handling consistent; change the joining method. Repeat specimens to see whether the result is consistent. A 100 g holding test does not find a join’s maximum strength.

Teacher answers are in the [separate guide](#).

Design brief | A small card pocket

Lesson 6 · Design for a user · part 1 of 2

This is a lightweight classroom prototype, not a wallet for valuables or a load-bearing bag. The design can be for a fictional club helper; no personal names or disclosures are needed.

USER AND PURPOSE: a club helper needs a small soft pocket to keep one paper task card together on the desk. It will be used with its opening upwards.

CARD: the test card is 60 mm wide × 50 mm high. It should slide in and come out without tearing.

SIZE: use one 100 mm × 170 mm fabric blank. The finished folded pocket is about 100 mm wide × 70 mm high. Two 10 mm-wide side joins leave about 80 mm internal width.

FUNCTION: after the appropriate joining/cure time, insert and remove the test card three times. Both side joins should stay closed. This is a desk-use check, not an upside-down shake or a carrying-strength test.

Teacher answers are in the [separate guide](#).

Design brief | A small card pocket

Lesson 6 · Design for a user · part 2 of 2

This is a lightweight classroom prototype, not a wallet for valuables or a load-bearing bag. The design can be for a fictional club helper; no personal names or disclosures are needed.

DESIGN: choose an opaque, flexible sample and a suitable cold join. Add a simple symbol or broad stripe on the front to distinguish the pocket. Do not use personal names, cords, sharp fasteners or loose decorations.

PROCESS: the plan must name fabric, join, dimensions, decoration position and ordered making steps. An adult checks the actual tape or adhesive and material before making.

PAPER/CARD ALTERNATIVE: use the same-size paper or light-card blank and school-approved cold tape. Record it as a model of the design, not a fabric-strength result.

Teacher answers are in the [separate guide](#).

Pattern and fold guide | Exact dimensions

Lesson 6 · Design for a user · part 1 of 2

Use the supplied full-size pattern at 100% or draw it with a ruler. Check the 50 mm scale line before cutting. Do not use “fit to page”. A teacher can trace or pre-cut the blank.

Blank: a rectangle 100 mm wide × 170 mm high. Keep the short edge at the top. Use millimetres; 100 mm = 10 cm and 170 mm = 17 cm.

Measure down from the top short edge. Mark a light horizontal fold line at 30 mm. Mark a second fold line at 100 mm. The top 30 mm is the flap; the middle 70 mm is the back; the bottom 70 mm becomes the front.

On the bottom/front panel, mark a 10 mm-wide joining strip at each side: left strip from $x = 0$ to 10 mm; right strip from $x = 90$ to 100 mm. Both run from $y = 100$ to 170 mm. Leave the middle of this panel free of adhesive.

For assembly, fold the bottom 70 mm panel upwards along the 100 mm line, matching its side edges to the middle/back panel. Press only the two side joining strips as the product instructions require. The top edge of the pocket must remain open.

Teacher answers are in the separate guide.

Pattern and fold guide | Exact dimensions

Lesson 6 · Design for a user · part 2 of 2

Use the supplied full-size pattern at 100% or draw it with a ruler. Check the 50 mm scale line before cutting. Do not use “fit to page”. A teacher can trace or pre-cut the blank.

Fold the top 30 mm flap down along the 30 mm line. The flap rests over the opening; it does not need a fastener. The closed outline is approximately 100 mm × 70 mm.

Test card: cut a separate rectangle 60 mm × 50 mm. Use paper or light card, with no confidential details. Slide it through the opening upwards and remove it gently.

Decoration zone: on the outside front after folding, draw one large symbol within an imaginary central 50 mm-wide × 35 mm-high area. Keep decoration away from side joins and the opening. Use a school-approved fabric marker; no added beads/buttons.

This pattern uses two side-join allowances of 10 mm. It is not a stitched seam pattern and should not be resized silently; a size change needs a fresh fit check.

Teacher answers are in the [separate guide](#).

Two original design ideas | Compare and change

Lesson 6 · Design for a user · part 1 of 2

These written idea cards replace an internet image search. Both use the supplied 100 mm × 170 mm blank. Borrow a design feature and change it for the stated user.

IDEA A - Signal pocket: opaque felt; two cold fabric-tape side joins; finished size about 100 mm × 70 mm. A large outlined triangle on the front helps the club helper identify it quickly.

IDEA B - Stripe pocket: the same pocket shape and side joins. Two broad drawn stripes make it look different from other pockets. The flap and opening stay clear.

USER CARD 1 - Alex, a fictional club helper, needs to spot the task-card pocket quickly on a busy desk. A large clear symbol matters more than tiny decoration.

Teacher answers are in the [separate guide](#).

Two original design ideas | Compare and change

Lesson 6 · Design for a user · part 2 of 2

These written idea cards replace an internet image search. Both use the supplied 100 mm × 170 mm blank. Borrow a design feature and change it for the stated user.

USER CARD 2 - Rowan, a fictional display helper, wants a quiet, simple design. One broad stripe is preferred, and the pocket must still accept the 60 mm × 50 mm card.

Design choices: choose one user, one specimen (or paper model), a tested cold join and a simple front symbol. Explain which feature came from an idea card and what you changed.

If a flat decoration is chosen instead of the pocket: use a 100 mm × 70 mm paper/card-backed fabric panel and a 60 mm × 50 mm front panel, joined with two 10 mm × 50 mm cold-tape strips at the front panel's side edges. It must stay flat on the desk; do not claim it can contain a card.

Teacher answers are in the [separate guide](#).

Plan-review checklist

Lesson 6 · Design for a user

One person explains the plan while the other points to each feature. Ask before altering someone else's drawing.

Can I identify the intended user and the item's limited purpose?

Can I find the chosen material and a relevant observed property?

Can I find the cold joining method and its positions? A pocket also needs a clear opening.

Can I read the dimensions and follow the pocket folds or flat-panel position?

Can I see the symbol/stripe position, clear of the joins and any pocket opening?

Can I follow the making order and locate adult approval plus required handling/cure time?

Can I find the test that will show whether the design meets its brief?

[Teacher answers are in the separate guide.](#)

Make the pocket | Complete construction card

Lesson 7 · Make, test and improve · part 1 of 2

Allow approximately 17 minutes for assembly using a prepared blank. An adult trials the chosen cold tape on the actual fabric before the lesson and checks its handling/cure instructions. If it cannot be tested within the session, use a pre-cured example or the paper model for evaluation.

YOU NEED: one 100 mm × 170 mm approved felt/fabric blank; two 10 mm × 70 mm strips of school-approved cold, pressure-applied fabric tape; a 60 mm × 50 mm paper test card; ruler; pencil/chalk; approved fabric marker. Adult-controlled scissors are optional because all parts can be pre-cut.

1. Read the design brief and your plan. Ask the adult to confirm the material, join and tools are approved. Choose maker, reader or checker roles.

2. Check the blank is 100 mm × 170 mm. With its short edge at the top, mark folds 30 mm and 100 mm down from that edge, or use the pre-marked blank.

3. Practise the folds before adding tape: bottom panel up, top flap down. Check the card can sit inside. Unfold so the joining side faces upwards.

4. Place the two 10 mm × 70 mm cold-tape strips on the left and right edges of the bottom panel only. They run from the 100 mm fold to the bottom edge. Ask the checker to confirm the centre and opening are clear.

5. Remove tape backing as directed, fold the bottom panel up along the 100 mm line and align side edges. Press the two side joins for the time and pressure stated on the product label. Keep the opening clear.

Teacher answers are in the [separate guide](#).

Make the pocket | Complete construction card

Lesson 7 · Make, test and improve · part 2 of 2

Allow approximately 17 minutes for assembly using a prepared blank. An adult trials the chosen cold tape on the actual fabric before the lesson and checks its handling/cure instructions. If it cannot be tested within the session, use a pre-cured example or the paper model for evaluation.

6. Fold the top flap down. Add the planned large symbol or stripe to the outside front using the approved marker. Avoid wet joins and allow any marker drying time required.

7. When the join is ready to handle according to its label, open the flap and gently insert/remove the 60 mm × 50 mm paper card three times. Check fit and both side joins. Record not yet tested if the join needs more cure time.

8. Improve one point if it can be done safely: align an open edge, replace a failed join under adult direction, or clarify the symbol. Repeat the relevant check; do not hide a fault with decoration.

9. Record one observed action and one test result. Store the item flat in the agreed place. Collect tape backings and offcuts; return counted tools and clear the workspace.

PAPER MODEL: use the same blank, folds and test card with suitable cold paper tape. Record a construction/fit model. Its result does not establish that a chosen fabric join will work.

FLAT-DECORATION ALTERNATIVE: use a 100 mm × 70 mm backing and a 60 mm × 50 mm front panel. Centre the smaller panel with 20 mm side margins and 10 mm top/bottom margins. Join its two sides using 10 mm × 50 mm cold-tape strips. Draw a symbol; after the required handling time, lift from the desk three times and check both panel joins remain attached.

Teacher answers are in the separate guide.

Prototype records | Fictional results

Lesson 7 · Make, test and improve

Use these complete paper records when practical testing is unavailable. They are invented examples, not evidence that a pupil made or tested an item.

BRIEF: pocket approximately 100 mm × 70 mm when folded; test card 60 mm × 50 mm fits; both side joins stay closed through three gentle insert/remove cycles; a clear front symbol; adult-approved materials and method.

POCKET A: finished 100 mm × 70 mm; card fits; joins held through all three cycles; a large triangle is visible; the adult-approved tape's required handling time was completed.

POCKET B: finished 100 mm × 70 mm; card fits at first, but the left join opens on cycle 2. A clear stripe is visible; tape had been correctly prepared and its handling time completed.

POCKET C: finished 100 mm × 70 mm; looks neatly folded, but no card test has happened because the join needs longer to cure. Material and tool checks are recorded; a symbol is planned but not added yet.

Possible verdicts: met = evidence shows the criterion worked; not yet = evidence shows it did not work; need evidence = the check has not happened or its result is missing.

Useful improvement statement: "The left join opened on cycle 2, so I will ask the adult to help replace the join, then repeat three gentle card cycles."

Teacher answers are in the separate guide.